

Hooks, Props and State



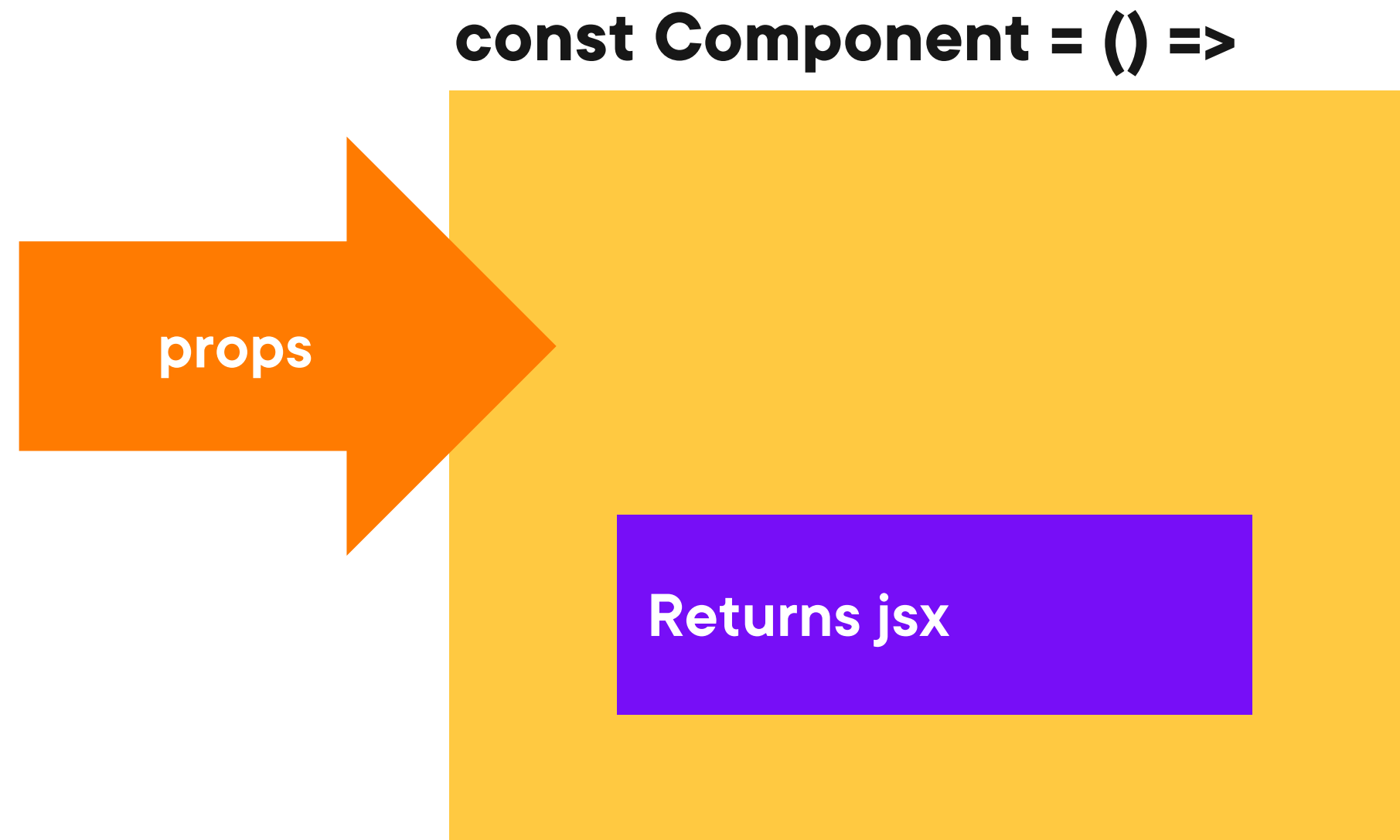
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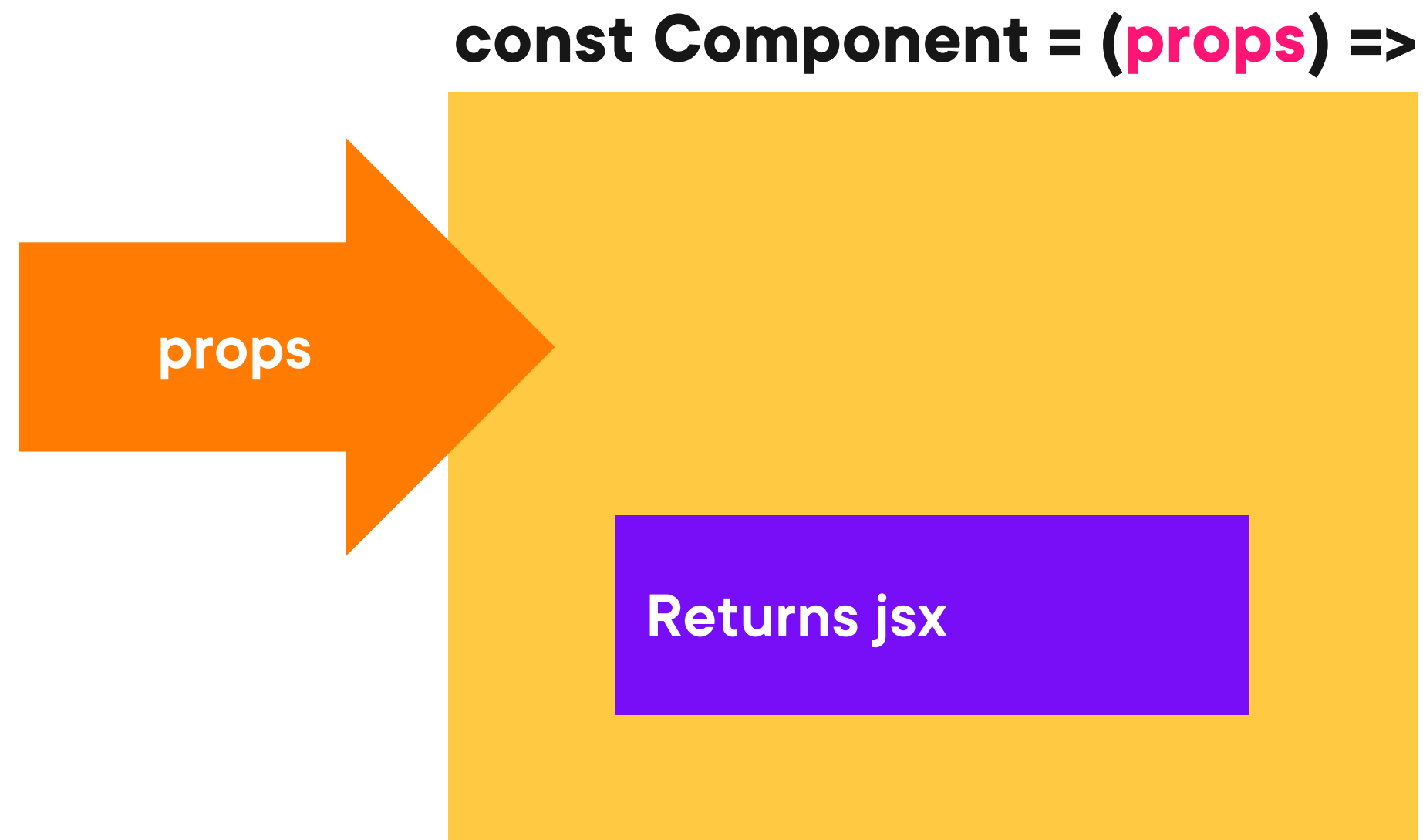
Props



<Component myprop="somevalue" />



Props



`<Component myprop="somevalue" />`



Rule: Props are read-only



Props Takeaways

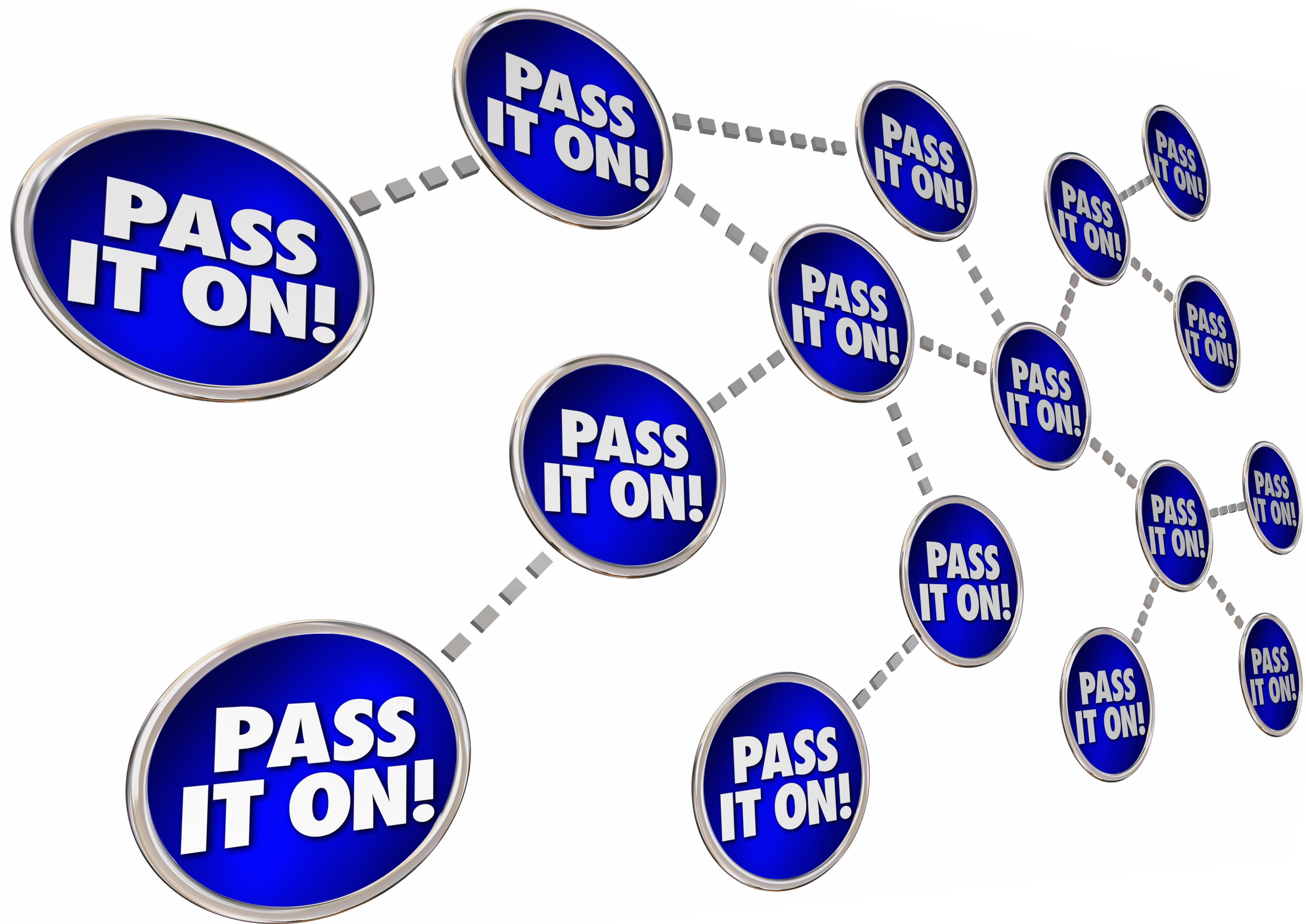
Passing arguments to components

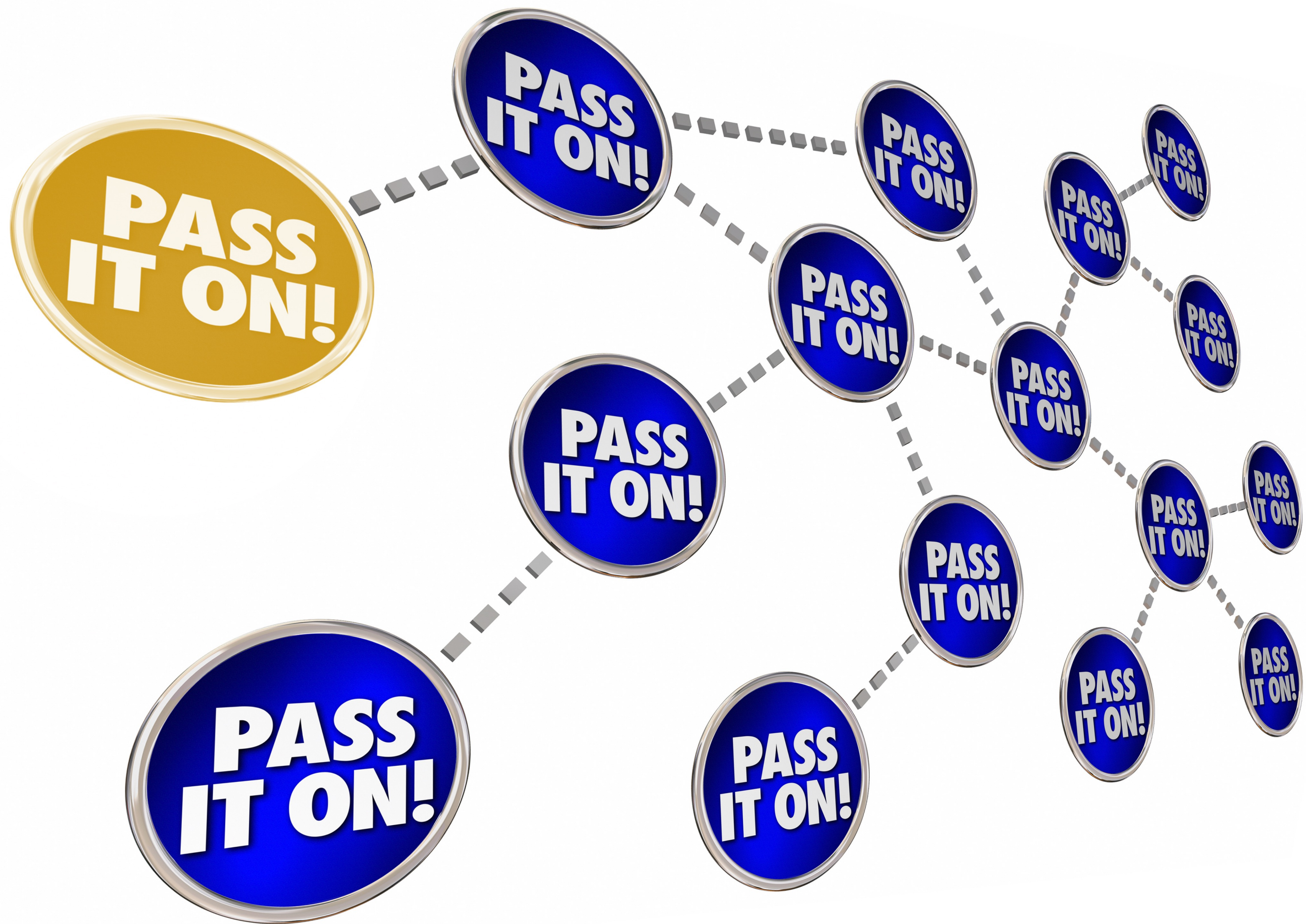
Using HTML-like attribute syntax

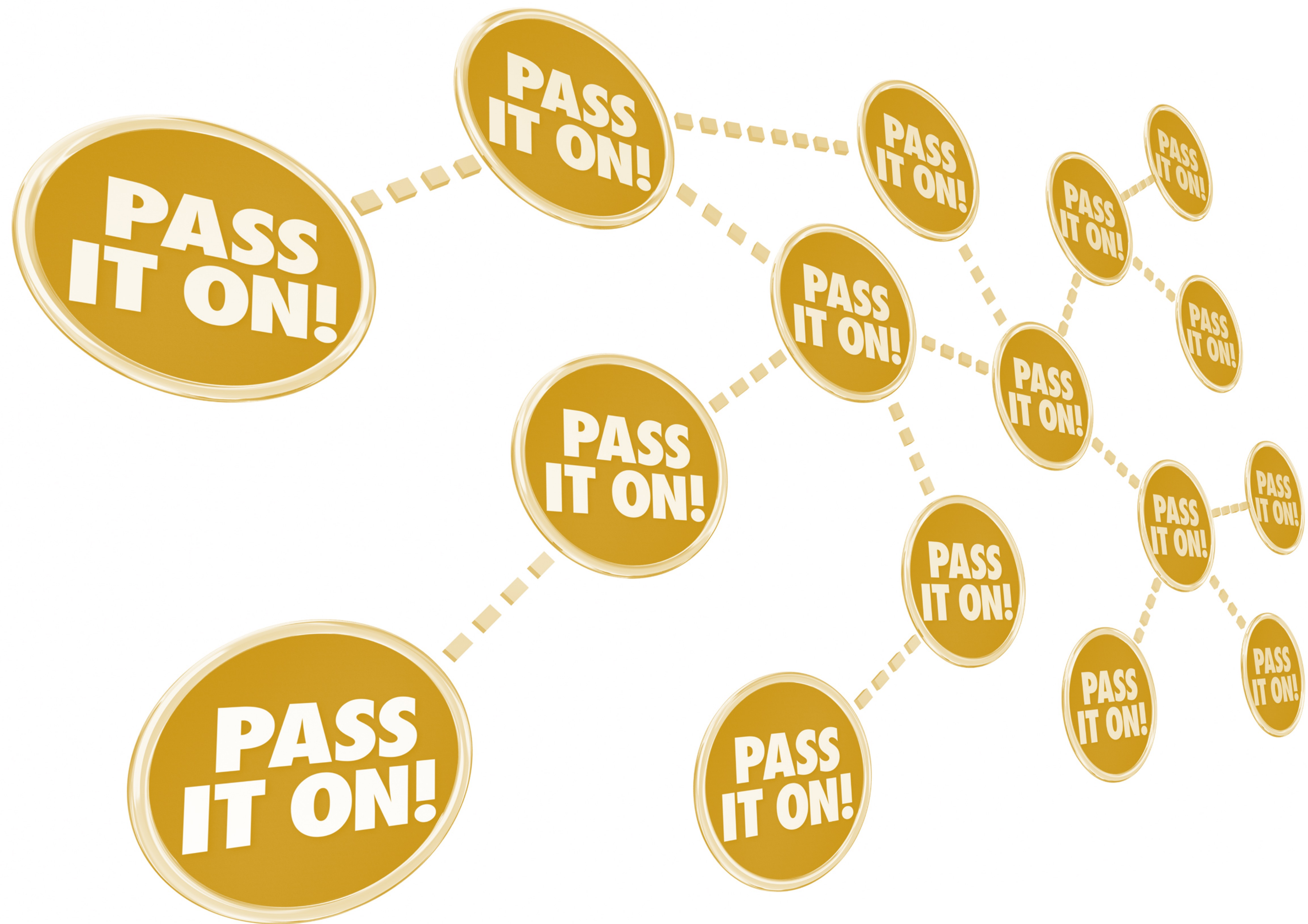
Receiving component accesses props object

Allow for better reusability of components



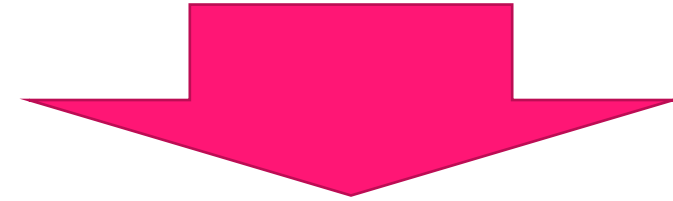




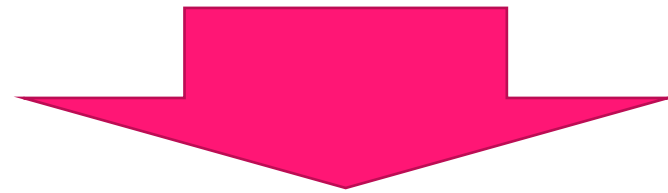


map

```
const numbers = ["one", "two", "four"];
```



```
const numbersPrefixed = numbers.map(n => "Number " + n);
```

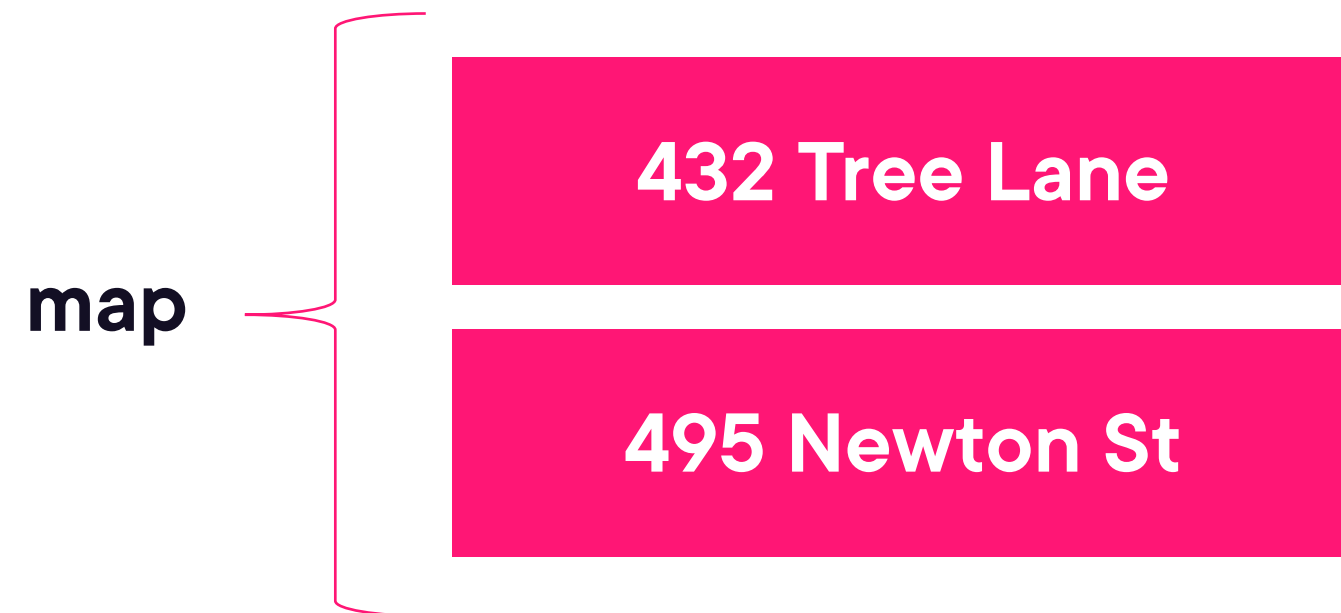


```
// ["Number one", "Number two", "Number four"]
```

The key prop

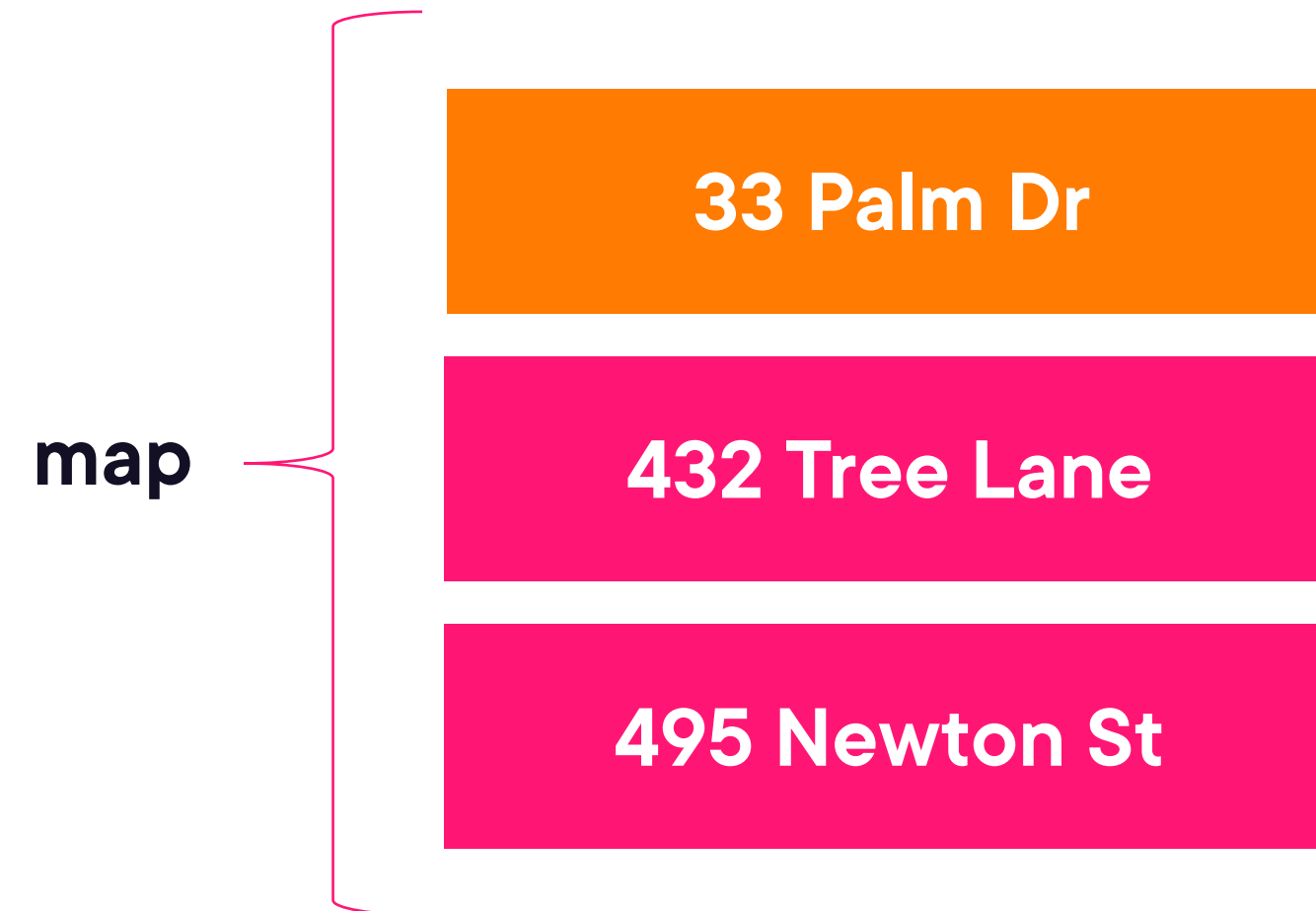
Array:

```
{ address: "432 Tree Lane" }  
{ address: "495 Newton St" }
```



Array:

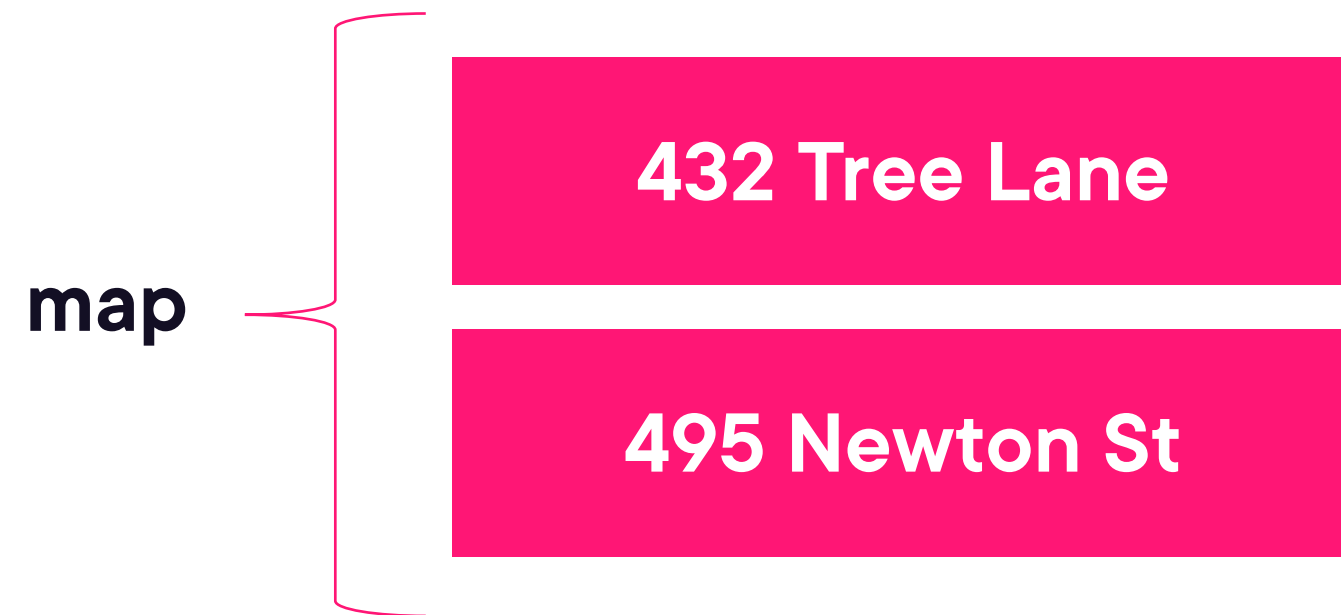
```
{ address: "33 Palm Dr" }  
{ address: "432 Tree Lane" }  
{ address: "495 Newton St" }
```



The key prop

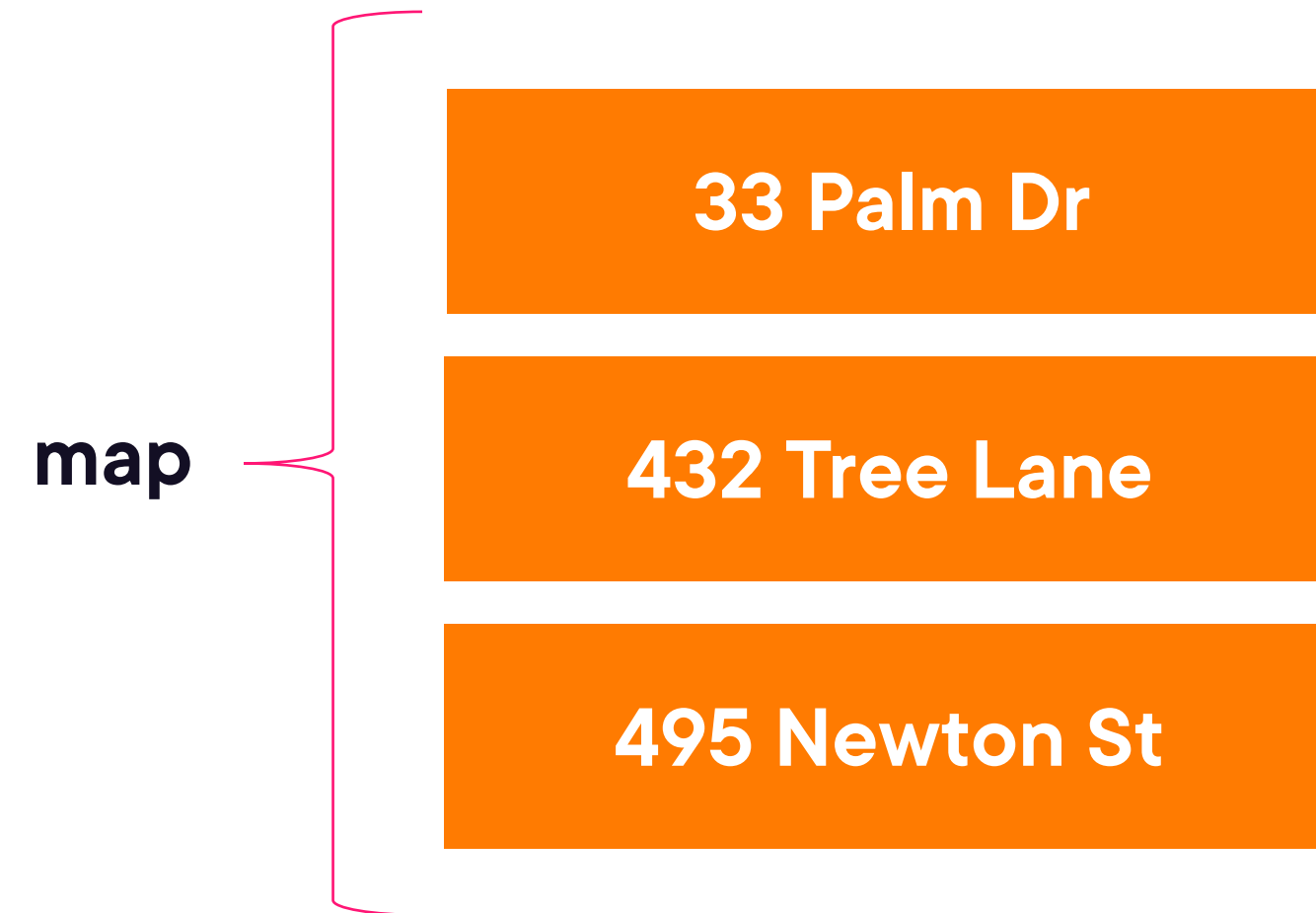
Array:

```
{ address: "432 Tree Lane" }  
{ address: "495 Newton St" }
```



Array:

```
{ address: "33 Palm Dr" }  
{ address: "432 Tree Lane" }  
{ address: "495 Newton St" }
```



The key prop

Array:

```
{ id: 1, address: "432 Tree Lane" }  
{ id: 2, address: "495 Newton St" }
```

map

1. 432 Tree Lane

2. 495 Newton St



Choosing a Key Value

Use id property if present

Else any combination of properties
that is unique

Last resort: index provided by map function



Hooks

A function

Starts with "use"

Encapsulates complexity

Access React's features

Custom hooks

Imported



Rules of Hooks

Only call hooks at the top level

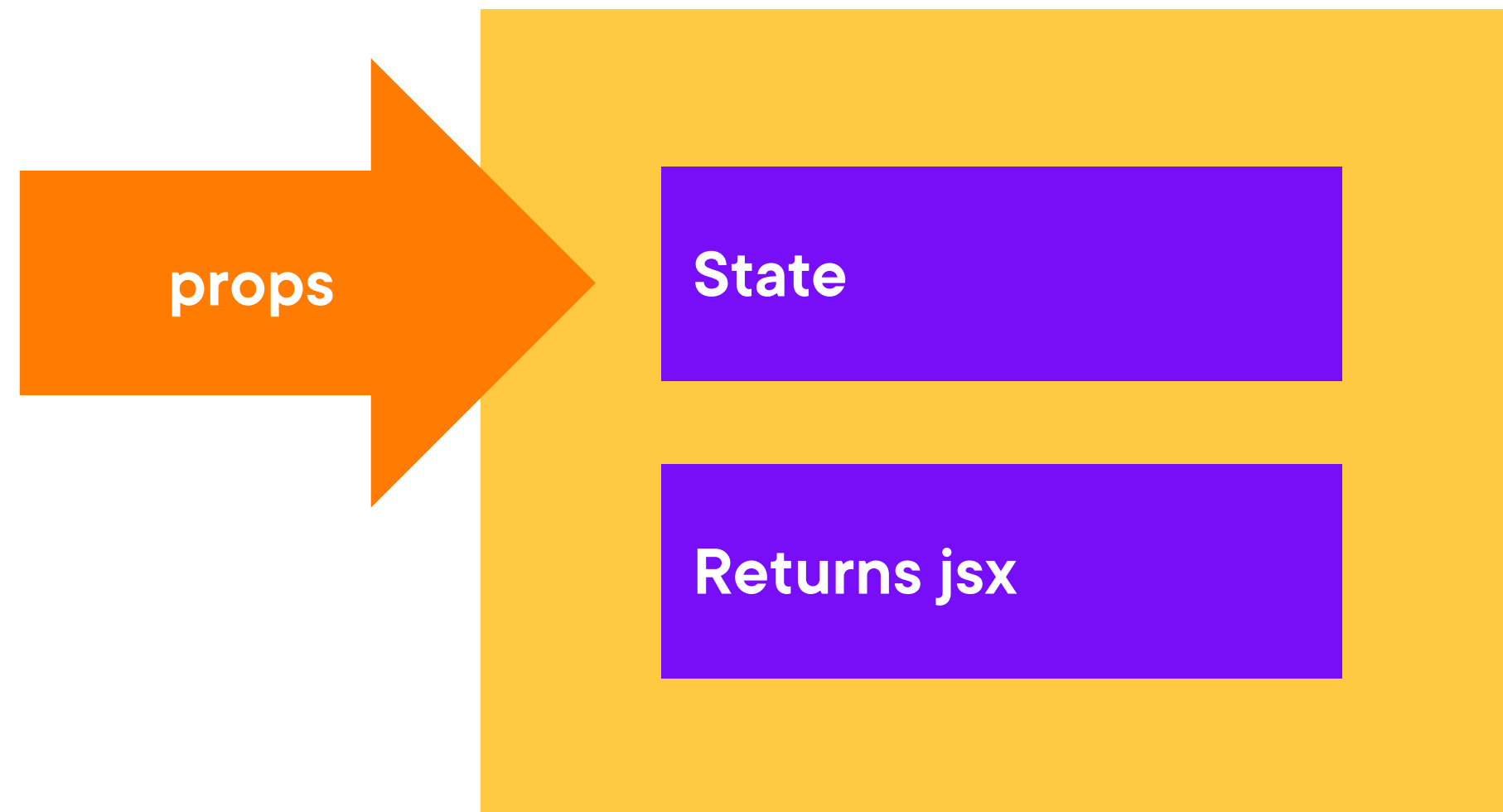
- They should always be called
- In the same order

Only call hooks in function components

- Or custom hooks

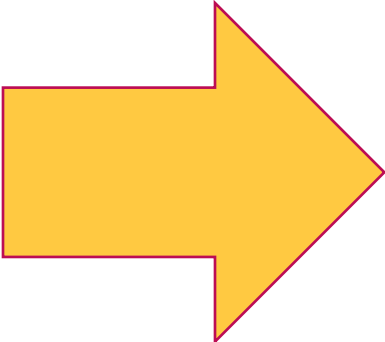


Components and State

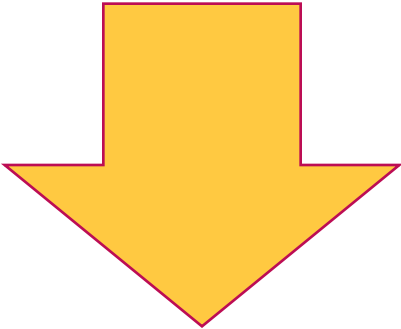


One Way Data Flow

```
[
  {
    id: 1,
    address: "12 Valley of Kings, Geneva",
    country: "Switzerland",
    price: 900000
  },
  {
    id: 2,
    address: "89 Road of Forks, Bern",
    country: "Switzerland",
    price: 500000
  },
]
```



```
{houses.map(h => (
  <HouseRow key={h.id} house={h} />
))}
```



Address	Country	Asking Price
12 Valley of Kings, Geneva	Switzerland	\$900,000
89 Road of Forks, Bern	Switzerland	\$500,000



State Main Takeaways

State is internal data managed by a component

Introduced by state hook

Parameter: initial value

Returns: array with current value and function to change it (the "set function")

Change the state using the set function only



A prop value can change



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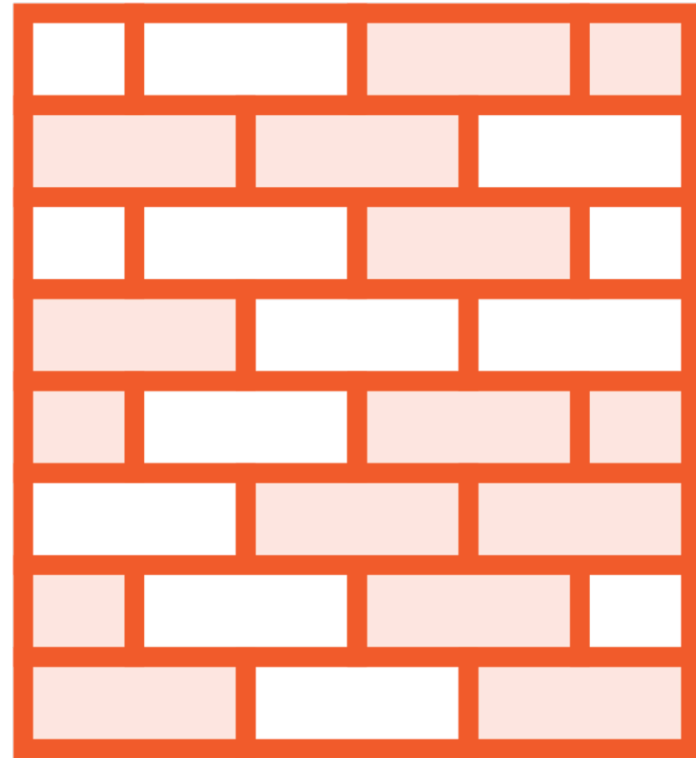
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Core React Features



Structure with components

Reusability

Have state



UIs declared in JavaScript

**Rendered output changes
when state is updated**



Efficiency with reconciliation

**Only updates the parts
of the UI that changed**

